

Neural SC Decoding of Polar Codes for the Two-User Gaussian MAC

5-Page Summary Report (March 2026)

1. Problem Statement

We develop a **neural network-based successive cancellation (SC) decoder** for polar codes on the two-user Gaussian Multiple Access Channel (GMAC). Two users transmit independently encoded polar codewords $X = F^n(U)$ and $Y = F^n(V)$ through $Z = (1-2X) + (1-2Y) + W$, where $W \sim N(0, \sigma^2)$. The receiver must decode both U and V .

The analytical SC decoder achieves excellent BLER by walking the polar code factor graph tree and computing exact transition probabilities. Our goal: replace ALL analytical operations with learned neural networks, enabling deployment on channels where analytical probabilities are unavailable.

Channels implemented: BEMAC ($Z=X+Y$, discrete), ABNMAC (binary noise), GaussianMAC (BPSK + AWGN).

Decoding paths: Three classes determined by the decoding order of U and V bits: - Class A/C (extreme paths): decode one user entirely before the other — simpler, no CalcParent needed - Class B (interleaved): alternating U and V decisions — hardest, requires CalcParent operations

2. Neural Decoder Architecture

The decoder replaces each analytical tree operation with a learned MLP:

Operation	Analytical	Neural	Input -> Output
Channel embedding	$W(z x,y)$	$z_encoder$ MLP	$R \rightarrow R^d$
CalcLeft (f-node)	Circ. conv.	MLP	$R^{\{3d\}} \rightarrow R^d$
CalcRight (g-node)	Cond. product	MLP	$R^{\{3d\}} \rightarrow R^d$
CalcParent	Inv. circ. conv.	Gated residual MLP	$R^{\{2d\}} \rightarrow R^d$
Decision	Prob. extraction	MLP	$R^d \rightarrow R^4$

Key design: All operations are **weight-shared** across tree positions and depths, making the model N -independent ($\sim 25K$ - $39K$ parameters). A model trained at $N=128$ can decode at any N .

CalcParent uses a gated residual structure: $output = gate * MLP(input) + (1-gate) * (left+right)/2$, providing stable gradient flow through deep trees.

Training: Sequential teacher-forced SC tree walk with cross-entropy loss on the 4-class joint (u,v) decisions. Curriculum learning ($N=16 \rightarrow 32 \rightarrow 64 \rightarrow 128 \rightarrow \dots$) is essential — from-scratch training fails at $N \geq 32$.

3. Key Results

BEMAC (solved)

The neural decoder **matches or beats** the analytical SC decoder at ALL N up to 1024: - N=64: NN-SC achieves 0.50x SC BLER (2x better) - Neural SCL(L=4) beats analytical SCL(L=4) by up to 8x at N=64

GMAC (partially solved)

N	SC BLER	NN-SC BLER	Ratio	Status
32	0.046	0.046	1.0x	Solved
64	0.025	0.026	1.03x	Solved
128	0.016	0.019	1.17x	Close
256	0.005	0.019	3.7x	Gap
512	0.001	0.045	40x	Large gap

N $\leq$ 128 is essentially solved. The gap at N $\geq$ 256 is the core open problem.

SCL Baselines (analytical, for comparison)

SCL(L=4) at GMAC 6dB: BLER = 0.026 (N=32), 0.013 (N=64), 0.008 (N=128), **0.0005 (N=256)**, 0.000 (N $\geq$ 512).

4. What Was Tried

Approaches that helped:

- **Stable cosine LR** (no warm restarts): Improved N=128 from 1.69x to 1.17x SC
- **Larger model (d=32, 157K params)**: Matches SC at N=32 (vs d=16's 1.22x), but under-trained at N $\geq$ 64
- **Per-level CalcLeft/CalcRight**: Separate MLPs per tree depth — solves curriculum transfer failure (d=32 model started at BLER=1.0 at N=128, per-level started at 0.265)
- **Neural SCL**: List decoding on NN output beats analytical SCL(L=4) at N $\leq$ 64

Approaches that failed:

- **NPD-style fast_ce (parallel training)**: Works for single-user binary, but 4-class MAC joint prediction plateaus at near-random loss. Two-decoder binary approach impossible for Class B ($R_u >$ marginal capacity).
- **d=32 at N $\geq$ 64**: Needs disproportionately more training, curriculum transfer fails entirely at N=128
- **Knowledge distillation variants, LLR front-end, FiLM conditioning**: Faster convergence but same BLER ceiling
- **BEMAC-to-GMAC weight transfer**: Works at N=32, fails at N $\geq$ 64

Root cause of $N \geq 256$ failure:

Weight-shared MLPs accumulate small per-node approximation errors through $\log_2(N)$ tree levels. At $N=256$ (8 levels), the accumulated error dominates. The sequential training's $O(N \log N)$ gradient depth exacerbates this by making it hard to learn fine-grained corrections at deep levels.

5. Open Problems and Directions

1. **Adapting NPD's fast_{ce} to MAC:** The single-user NPD uses binary sign-flips that match the analytical g-node structure. The MAC's 2x2 circular convolution doesn't factor this way. A correct parallel training formulation for the joint MAC decoder is the highest-impact unsolved problem.
2. **Residual connections in CalcLeft/CalcRight:** NPD's BitNode uses `output = MLP(...)` + `analytical_approximation`. Our CalcLeft/CalcRight have no residual structure. Adding one could stabilize training through deep trees.
3. **Per-level with long training:** Per-level ops solve curriculum transfer. Combined with stable LR and sufficient budget (days), this could close the $N=256$ gap.
4. **Architecture search:** Transformer-based tree operations, attention across levels, or other architectures that don't accumulate errors multiplicatively.
5. **GA design is wrong for Class B:** Must use MC design. This was a critical bug that went unnoticed for a long time.

Code and data: <https://github.com/eitanspi/polar-codes-mac>